



Ornithopters based on Arduino board and dash receiver

Team: Golden Bird

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Introduction

Design Background:

The ornithopter is an aircraft whose wings flat up and down like the wings of birds and insects, which is usually applied in military applications, environmental detection and sampling, agriculture, scientific research.

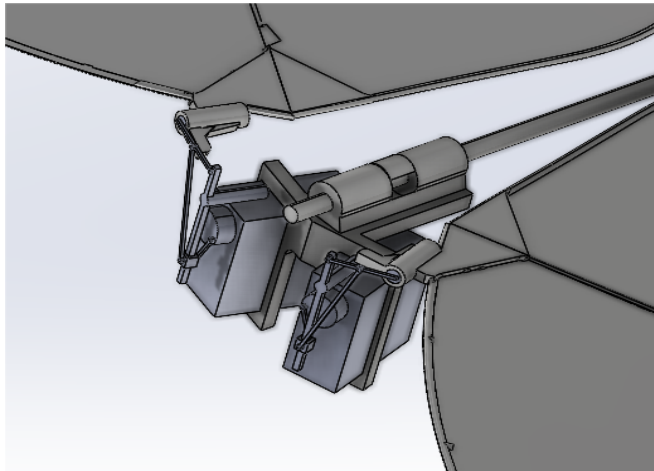
Products information:

Our group design two versions of ornithopters. One is powered by servos and remotely controlled by an Arduino board. The other is powered by brushless motor and controlled by dash receiver. And they follow different principles.

Methodology

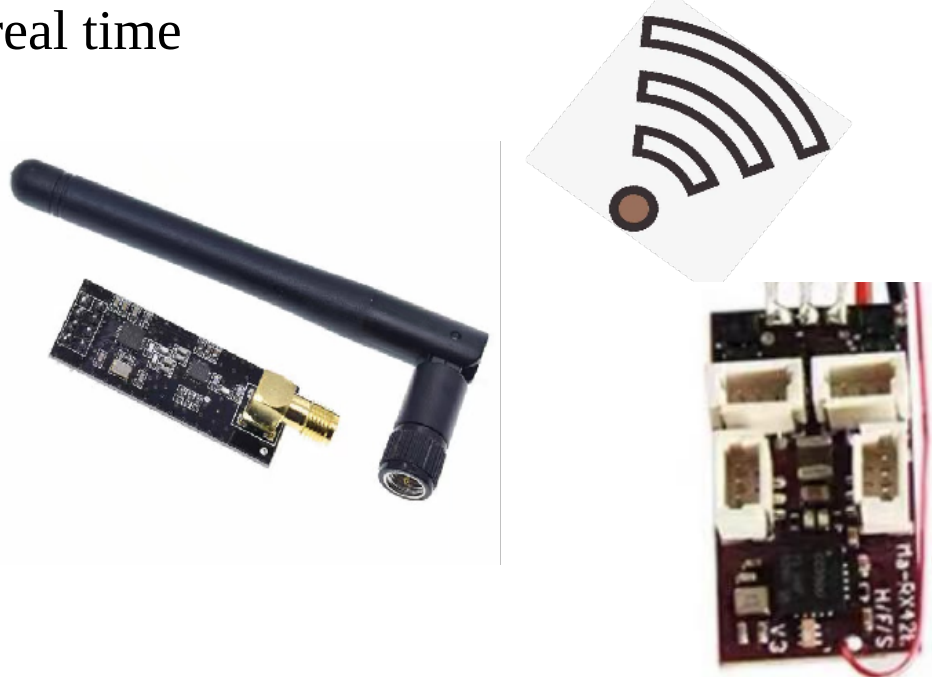
To make the ornithopter can change directions during the flight:

We use two servos on both sides. If one servo works at a higher frequency under the control of arduino board which contains related code, the ornithopter will turn to the other way.



To realize the remote control function:

We use antenna and dash receiver connected so that we could control it in real time



To make the ornithopter fly steadily:

We consider different structure of creature like birds, butterflies and dragonflies.

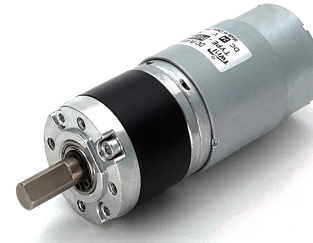


Procedure and Results

Materials used:

Version I

Carbon plate
Carbon fiber rod
DS 1906B steering engine
Arduino nano board
Motive power lithium Battery
Kite cloth
3D printed parts
NRF24L01 wireless module adapter board
NRF24L01+PA+LNA wireless module
Transmit and receive module



Version II

X6 FUTABA S-FHSS remote control
Brushless motor OP03X 20000KV
Ultra micro low voltage digital steering

Problems and issues:

1. Can't use SolidWorks fluently
2. Materials are difficult to assemble
3. Some parts sometimes break down
4. Structure is complex
5. Code is hard to learn

Experiment procedure:

We first looked at the papers, settled on a model of the ornithopter, built a 3D model with SolidWorks, then bought the required parts and 3D printouts, learned how to write the code, and finally put them together. And we built the other one, which repeated the same procedure but with different materials.

Results:

The ornithopter achieves our goals and it can perfectly complete the task



Version I

Version II



Discussion and Analysis

In the process of making ornithopters, we deeply felt the charm of scientific and technological innovation. From the initial idea to the final assembly, every step is challenging but fun.

During the production process, we encountered some difficulties, such as choosing the right materials, designing an efficient flight mechanism, etc. However, we did not give up. Instead, we actively researched and tried to find a solution to the problem.

Conclusions

After completing this project, we felt a great sense of accomplishment

In short, the process of making ornithopters made us feel the charm, challenge and fun of technological innovation.

We believe that this experience will have a positive impact on my future studies and work.

Acknowledgement

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References



<https://www.arduino.cc>
<https://github.com/>
<https://www.csdn.net>